

PERRY, IA, USA

WMO: 720412

Lat: 41.828N

Lon: 94.160W

Elev: 309

StdP: 97.67

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 54943

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.0	-18.6	-25.9	0.4	-21.5	-22.6	0.5	-18.1	13.7	-3.9	12.4	-5.6	3.5	310	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	33.8	22.7	32.3	23.0	30.7	22.5	26.1	30.3	25.0	29.4	24.0	28.2	5.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	20.9	28.9	23.8	19.4	27.3	22.8	18.2	26.3	82.8	30.3	78.0	29.2	73.7	28.1	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
11.8	10.5	9.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	-27.0	27.9	3.3	1.3	-29.4	28.9	-31.3	29.6	-33.2	30.4	-35.6	31.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-5.9	-5.1	3.2	10.2	17.0	22.2	23.0	21.8	18.7	11.3	3.5	-3.1
	DBStd	11.75	7.01	6.75	6.45	5.29	5.11	3.53	3.37	2.89	4.58	5.07	5.79	6.19
	HDD10.0	1872	494	423	228	61	7	1	0	0	2	46	205	406
	HDD18.3	3612	753	656	471	247	87	8	3	4	49	226	445	664
	CDD10.0	1800	0	1	17	67	222	367	403	365	263	85	11	0
	CDD18.3	498	0	0	1	4	44	124	148	111	59	6	0	0
	CDH23.3	4859	0	0	11	73	458	1199	1481	957	597	80	2	0
	CDH26.7	1754	0	0	1	18	167	459	582	302	207	18	0	0
(14) Wind	WSAvg	4.0	4.7	4.5	4.8	5.2	4.7	3.9	2.7	2.4	2.9	3.7	4.5	4.5
(15) Precipitation	PrecAvg	911	27	27	55	93	124	136	123	110	78	61	47	30
	PrecMax	1176	51	48	130	203	232	288	287	226	166	178	116	64
	PrecMin	603	10	3	2	18	36	17	32	21	26	9	2	0
	PrecStd	158	11	13	33	49	51	71	67	59	40	45	32	18
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	16.1	25.0	29.5	34.4	35.1	37.0	34.3	33.7	29.1	22.7	13.5
		MCWB	6.7	11.7	17.8	18.5	18.8	23.7	24.2	22.8	21.4	17.6	13.5	8.6
	2%	DB	7.6	9.9	20.7	25.2	31.0	32.9	33.8	32.0	31.4	25.4	17.9	9.9
		MCWB	5.0	7.2	13.9	15.1	19.6	22.6	24.4	23.2	22.6	15.9	11.7	7.3
	5%	DB	4.9	6.4	17.0	22.3	27.9	31.3	32.1	29.8	29.0	22.7	15.0	7.3
		MCWB	3.1	4.4	12.9	13.5	18.0	22.1	24.9	23.3	20.9	15.4	9.9	4.9
	10%	DB	2.8	3.2	12.9	19.9	25.8	29.5	30.0	27.9	27.0	20.2	12.4	4.6
		MCWB	1.6	2.0	9.2	12.5	17.2	21.1	23.3	22.0	19.5	14.1	8.5	3.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.4	11.4	19.3	19.6	23.0	26.8	28.3	26.8	25.1	21.1	15.2	10.8
		MCDB	10.4	15.3	24.2	24.1	29.0	32.1	32.1	30.6	30.5	23.2	19.2	11.3
	2%	WB	5.1	7.5	15.8	17.0	21.3	25.1	26.7	25.2	24.0	19.0	13.1	7.4
		MCDB	7.5	9.5	18.9	22.1	26.6	30.0	30.6	28.7	28.6	22.9	17.9	9.4
	5%	WB	3.3	4.4	12.7	15.3	20.2	23.9	25.4	24.1	22.9	17.1	10.7	5.1
		MCDB	4.8	6.4	16.1	21.0	25.2	28.5	29.8	27.0	26.5	20.1	13.8	6.9
	10%	WB	1.7	2.2	9.7	13.6	19.0	22.8	24.4	23.0	21.6	15.2	8.5	3.1
		MCDB	2.7	3.4	13.1	18.2	24.0	27.4	28.9	26.9	25.0	19.0	12.3	4.6
(35) Mean Daily Temperature Range	MDBR		9.8	9.5	9.9	12.1	11.6	11.1	11.5	11.4	13.3	12.5	10.7	8.9
	5% DB	MCDBR	12.2	11.7	16.1	17.2	14.8	13.1	13.0	13.1	14.3	15.9	15.3	12.1
		MCWBR	9.5	8.9	9.9	8.9	5.7	5.7	6.1	6.4	6.1	7.9	9.0	9.1
	5% WB	MCDBR	11.3	11.7	14.3	13.7	11.7	11.2	11.6	11.0	11.5	12.3	13.6	10.7
(40) Clear-Sky Solar Irradiance	taub		0.293	0.308	0.327	0.382	0.411	0.421	0.435	0.422	0.388	0.342	0.298	0.289
	taud		2.405	2.370	2.410	2.276	2.266	2.284	2.227	2.274	2.321	2.442	2.504	2.437
	Ebn at Noon		857	899	922	887	866	855	838	839	842	848	849	831
	Edh at Noon		79	96	104	128	133	131	137	127	113	89	72	71
(44) All-Sky Solar Radiation	RadAvg		1.93	2.94	3.80	4.77	5.63	6.26	6.35	5.47	4.55	3.08	2.10	1.60
	RadStd		0.18	0.20	0.37	0.38	0.43	0.56	0.44	0.37	0.33	0.37	0.16	0.14

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page